1)

SELECT c.CustomerID, c.CustomerName, o.OrderID FROM Customer c INNER JOIN Orders o ON c.CustomerID = o.CustomerID;

2)

```
SELECT EmployeeName FROM Employee e
WHERE Salary >
(
   SELECT AVG(Salary)
   FROM Employee
   WHERE DepartmentID = e.DepartmentID
);
```

3)

CREATE VIEW EmpView AS SELECT DepartmentID, COUNT(*) AS TotalEmployees, AVG(Salary) AS AvgSalary FROM Employee xGROUP BY DepartmentID;

4)

Employee ⋈ Employee.DepartmentID = Department.DepartmentID Department

5)

```
CREATE TABLE Orders (
   OrderID INT PRIMARY KEY,
   CustomerID INT,
   FOREIGN KEY (CustomerID)
   REFERENCES Customer(CustomerID)
);
```

```sql
INSERT INTO Customer(CustomerID,
CustomerName) VALUES (500, 'John');

INSERT INTO Orders(OrderID,
CustomerID) VALUES (101, 500);
```

6)

```sql
SELECT e.EmployeeName
FROM Employee e
JOIN Department d
ON e.DepartmentID = d.DepartmentID
WHERE d.Location = 'Chennai';
```

7)

```sql
SELECT DepartmentID, COUNT(*) AS TotalEmployees  FROM Employee
GROUP BY DepartmentID;
```

8)

```sql
UPDATE Employee SET Salary = Salary + 5000 WHERE EmployeeID = 101;
```

9)

```sql
SELECT e.EmployeeName
FROM Employee e
JOIN Department d
ON e.DepartmentID = d.DepartmentID
JOIN Manager m
ON d.ManagerID = m.ManagerID
WHERE m.Experience > 10;
```

10)

$\{E \mid \exists D\ (Department(D) \wedge E.DepartmentID = D.DepartmentID)\}$